

Sparse Gaussian–Markov Modeling for Robust and Trustworthy Unknown Cyber Defense

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Abstract—Network intrusion detection systems (NIDS) are essential for enhancing network security by detecting and responding to attacks promptly. NIDS based on artificial intelligence (AI) offer better adaptability to unknown (out-of-distribution (OOD)) attacks compared to traditional anomaly detection methods. However, many existing AI-based NIDS directly adopt methods from other domains without adequately considering OOD data, which often leads to severe overfitting in trained classifiers and insufficient model reliability and robustness. This issue largely arises because most approaches focus solely on the relationship between individual features and predictions while neglecting the interactions among features, resulting in overfitting and overconfident predictions. Moreover, the retention of certain weak parameters may cause the model to maintain artificially high confidence when confronted with OOD data. To address this, we propose a Gaussian Markov Random Field that better captures traffic data distribution and feature correlations. Furthermore, our GMRF-based classification model uses sparsification with regularization L_0 to remove weak parameters. L_0 regularization counts the number of non-zero parameters and penalizes model complexity, thereby forcing insignificant weights to become exactly zero and improving generalization. This prevents high prediction probabilities for unknown attacks and effectively distinguishes them from benign traffic. Finally, we evaluated our method in the CICIDS-2017 and CSE-CICIDS-2018 datasets, achieving recall rates of 57% and 77% for unknown attacks while maintaining a recall of more than 95% for known traffic. Our approach significantly outperforms existing SOTA methods, and performance tests on a low-specification machine show a low detection latency of approximately 130 microseconds, making it suitable for real-time applications.

Index Terms—Unknown attacks, out-of-distribution, intrusion detection, GMRF, sparsification, real time.

I. INTRODUCTION

THE rapid development of the Internet has led to an explosion of cyber attacks, resulting in significant information and financial losses in critical sectors such as national defense, power grids, transportation and healthcare systems. In this context, NIDS have become essential components to ensure the security of modern information infrastructures. Traditional IDS approaches, which are based mainly on predefined rules and signatures, such as deep packet inspection [1], [2], [3], [4], [5], tend to be ineffective against previously unseen threats [6], [7]. As a result, recent research efforts have increasingly shifted toward AI-driven paradigms for intrusion detection [8], [9], [10], [11].

A. Existing AI-Based Work

As AI technologies continue to achieve remarkable success in various domains [12], [13], [14], their empowering role in intrusion detection has become increasingly prominent. Compared to traditional methods, AI-based approaches offer a distinct advantage in identifying unknown threats - those that lie beyond the distribution of in-distribution (ID) data [9], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29]. These emerging attack types often masquerade as normal traffic, evading detection by intrusion detection models, and thereby achieving malicious objectives. Existing research on detecting such unknown threats falls mainly into the following categories:

1) *Unsupervised or Semi-Supervised Methods*: In recent years, some researchers have explored reconstruction-based approaches to identify unknown threats [10], [15], [16], [17], [18]. These methods can be broadly categorized into two types. The first involves encoding ID data, often combined with a classification head to optimize classification performance and extract semantic representations of ID samples. This is followed by a reconstruction phase using a decoder. The second type trains directly an encoder-decoder framework to reconstruct ID data. In both cases, the detection of unknown threats is based on the reconstruction error: since out-of-distribution (OOD) samples are not involved in training, they typically exhibit higher reconstruction errors compared to ID samples.

Although this approach has achieved notable success in other domains, its direct application to the security domain remains limited due to domain-specific challenges. First, the

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data used for training in this context often consists of highly sparse network flow features. This sparsity significantly undermines the effectiveness of cosine similarity (as we demonstrate in *Appendix A*), which is commonly used as both an optimization objective and a detection metric. Second, due to the high similarity between benign and malicious flows, and the lack of genuine negative (malicious) samples during reconstruction training, the reconstruction-based mechanism often struggles to distinguish unknown threats effectively.

2) *Supervised Methods*: This category of methods uses traditional machine learning (ML) algorithms, such as decision trees, random forests, and support vector machines, as well as deep learning models, including multilayer perceptrons (MLPs), auto-encoders, stacked encoders, graph neural networks, and transformers, to build intrusion classification models using ID data [5], [20], [22], [24], [25], [30], [31], [32], [33], [34]. In addition, strategies such as contrastive learning and transfer learning have also been applied to NIDS research [21], [35], [36]. Despite their success in classifying known threats, these methods face two major limitations when encountering unknown threats.

Firstly, existing machine learning applications in NIDS often adopt standard algorithms directly without incorporating the specific context of cybersecurity. That is, they typically overlook the joint effects among features, the semantic relationships between attributes, and their associations with different types of attacks. This can lead to a high risk of overfitting during training. Secondly, since models are trained solely on ID data, they tend to learn superficial patterns of known traffic. Meanwhile, unknown threats are often designed to closely mimic benign traffic [15], [26], [37]. Although they fall outside the training distribution (i.e., they are OOD), they can still evade detection by these classifiers, further exacerbating the overfitting issue and undermining the model's ability to generalize to novel attack types.

B. Our Work

Our research is based on supervised learning methods and proposes a novel intrusion detection system, named ZeroXpert, which is specifically designed to enhance model reliability while mitigating overfitting. ZeroXpert utilizes the Gaussian Markov Random Field (GMRF) [38], [39] and sparsification with regularization L_0 , effectively captures spatial correlations between feature vectors, helping to reduce the risk of overfitting the model. Additionally, by applying sparsification to the model, we further decrease its dependence on weakly correlated features, thereby lowering the prediction probability when encountering unknown attacks. This approach enables us to identify unknown attacks by setting a threshold on the prediction probability. Finally, we conducted extensive experiments on two public datasets, CICIDS-2017 [40], [41] and CSE-CICIDS-2018 [41], [42]. Our method achieves recall rates of 55% and 77% for unknown attacks while maintaining a recall of more than 95% for known attacks and low false positive rates (around 2% and 4%) for benign traffic, demonstrating strong practical applicability. Compared with existing SOTA methods, ZeroXpert delivers superior performance across multiple metrics. In addition, detection latency

tests show an average of about 130 microseconds, highlighting its real-time capability.

In summary, our contributions to this article are as follows:

- We propose a novel approach using a pre-trained GMRF model for fine-grained classification of known data, which captures feature interactions to reduce overfitting and enhance robustness, improving detection of unseen attacks.
- We introduce a L_0 regularized parameter sparsification method built on a pre-trained GMRF model. Reduces reliance on weakly correlated features while preserving accuracy, further enhancing robustness in distinguishing unknown attacks through the prediction distribution.
- We evaluated our approach in the CICIDS-2017 and CSE-CICIDS-2018 datasets, achieving recall rates of 57% and 77% for unknown attacks and more than 95% for known traffic, surpassing several SOTA methods. On low-spec hardware, the detection latency is around 130 microseconds, confirming real-time applicability.

The remainder of the paper is organized as follows. Section II introduces the background of GMRF. Section III presents the detailed designs of our method. In Section IV and Section V, we describe the implementation of GMRF and sparsification, respectively. In Section VI, we experimentally evaluate the performance. Section VII reviews related work. Section VIII discusses our proposed method and further explores some potential future work directions. Finally, we conclude this paper in Section IX.

II. BACKGROUND

In this section, we detail the theoretical foundations of the Gaussian Markov Random Field [38], [39] to better address the two challenges we face:

- We should consider the effect of feature interactions on intrusion detection predictions.
- We need to minimize the impact of weakly correlated features on the prediction results.

GMRF is a probabilistic graphical model that is used to represent a set of random variables with Gaussian distributions, where the relationships between these variables satisfy the Markov property. Specifically, given the neighborhood of a random variable, the remaining variables are conditionally independent of the neighborhood of that random variable. This Markov property can be represented by a graph, where the edges indicate the conditional dependencies between variables.

Assume that a random feature vector X is represented as $[x_1, x_2, x_3, \dots, x_d]$. Its joint probability distribution $P(X)$ follows a multivariate Gaussian distribution, which can be expressed as:

$$P(X) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left[-\frac{1}{2} (X - \mu)^T \Sigma^{-1} (X - \mu) \right], \quad (1)$$

where μ is the mean vector, Σ is the covariance matrix, $|\Sigma|$ denotes the determinant of the covariance matrix.

Since $\frac{1}{(2\pi)^{\frac{d}{2}}|\Sigma|^{\frac{1}{2}}}$ is the model parameter and is independent of X , the probability distribution $P(X)$ can be expressed as follows:

$$\begin{aligned} P(X) &\propto \exp \left[-\frac{1}{2}(X - \mu)^T \Omega (X - \mu) \right] \\ &\propto \exp \left[-\frac{1}{2}(X^T \Omega - \mu^T \Omega)(X - \mu) \right] \\ &\propto \exp \left[-\frac{1}{2}(X^T \Omega X - X^T \Omega \mu - \mu^T \Omega X + \mu^T \Omega \mu) \right], \end{aligned} \quad (2)$$

where Ω represents the precision matrix, $\Omega = \Sigma^{-1}$.

Consider X and μ as a column vector of shape $[d, 1]$. Thus,

$$\begin{aligned} X^T \Omega \mu &= \mu^T \Omega X \\ P(X) &\propto \exp \left[-\frac{1}{2}(X^T \Omega X - 2\mu^T \Omega X + \mu^T \Omega \mu) \right]. \end{aligned} \quad (3)$$

Similarly, $\mu^T \Omega \mu$ is independent of X , and Ω is a real symmetric matrix, i.e. $\Omega = \Omega^T$, then

$$\begin{aligned} \mu^T \Omega^T &= (\Omega \mu)^T \\ P(X) &\propto \exp \left[-\frac{1}{2}(X^T \Omega X - 2(\Omega \mu)^T X) \right]. \end{aligned} \quad (4)$$

We analyze the density distribution of the flow features of the network. Taking flow duration as an example, in reality, most users' network flows end in a short time, with fewer users maintaining a long-duration network flow. After Min-Max normalization, the flow duration value density distribution approximates a normal distribution with a mean of 0. The features we extracted include time-related and packet length-related features, and our analysis shows that they conform to these characteristics.

Furthermore, we visualize the data distribution (randomly selecting six characteristics) in the CICIDS-2017 [40], [41] and CSE-CICIDS-2018 [41], [42] datasets in Fig. 1, and the results are consistent with our analysis. Based on the above analysis, we can set μ in Eq. (4) to a zero vector; thus,

$$\begin{aligned} P(X) &\propto \exp \left[-\frac{1}{2}(X^T \Omega X) \right] \\ X^T \Omega X &= [x_1, x_2, \dots, x_d] \begin{bmatrix} \lambda_{11} & \lambda_{12} & \dots & \lambda_{1d} \\ \lambda_{21} & \lambda_{22} & \dots & \lambda_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{d1} & \lambda_{d2} & \dots & \lambda_{dd} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \\ &= \sum_{i=1}^d \sum_{j=1}^d x_i x_j \lambda_{ij} \\ &= \sum_{i=1}^d \sum_{j=1}^d X X^T \odot \Omega, \end{aligned} \quad (6)$$

where $\odot$ represents element-wise multiplication.

We can obtain the expected objective:

$$P(X) \propto \exp \left[-\frac{1}{2} \left(\sum_{i=1}^d \sum_{j=1}^d X X^T \odot \Omega \right) \right]. \quad (7)$$

Ultimately, GMRF can be represented by a graphical model that describes the joint probability distribution of the nodes in

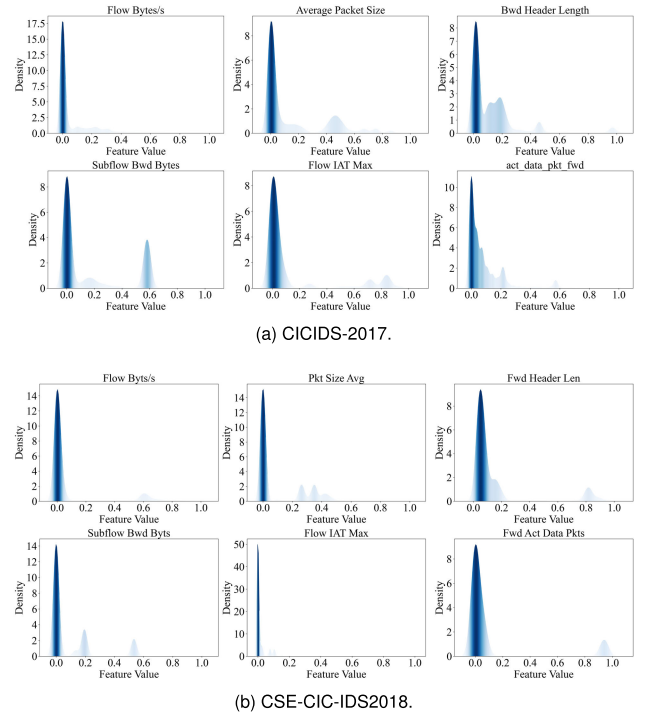


Fig. 1. The density distribution of CICIDS-2017 and CSE-CICIDS-2018 datasets. The x-axis represents the feature value, while the y-axis indicates the density, and darker colors correspond to higher density.

an undirected graph through cliques and potential functions. In this graphical model, each variable is represented as a node, and conditional independence relationships are represented as undirected edges, forming an undirected graph. In this model, cliques are fully connected subgraphs, and potential functions define the joint distribution of these cliques, thereby capturing dependencies between variables (as shown in Eq. (6), if $\lambda_{ij} = 0$, it indicates that there is no direct edge connecting the two nodes, meaning that they are independent). These potential functions allow the GMRF to flexibly represent complex probability distributions, effectively capturing local interactions and spatial dependencies.

Compared with traditional graph-based NIDS methods, GMRF does not construct graphs based on IP nodes, but instead models the relationships among features. Graph construction based on IP nodes faces a fundamental challenge: when the number of IP nodes is large, the resulting graph becomes extremely large, and the construction and computation overhead grow rapidly, making such models difficult to deploy in practice. In contrast, GMRF builds the graph over features, enabling the model to capture local interactions and spatial dependencies among features, which contributes to improved robustness. Traditional models tend to overfit to a few dominant features, whereas GMRF emphasizes the joint effects of multiple features, which is a key reason it can effectively alleviate overfitting. Moreover, our approach does not rely solely on GMRF. We aim to address these issues through the combination of structural constraints and parameter constraints. The former is imposed by GMRF-based feature dependency modeling, while the latter is achieved through parameter sparsification (pruning).

III. OVERVIEW OF OUR METHOD

In this section, we develop a real-time NIDS, named ZeroXpert, aimed at accurately predicting known traffic patterns while improving the detection capability for unknown attacks.

A. Design Philosophy

Unknown attacks may locally resemble benign traffic, making it easy to evade existing NIDS and be classified as benign by current AI models with a high prediction probability. We consider this phenomenon an issue of overconfidence in AI models and attribute it to their neglect of extensive spatial relationships between features, which refer to the combined effects of multiple features on model predictions.

To solve the above issues, we propose using a GMRF to learn the distribution of traffic features, with the aim of strengthening the learning of spatial relationships among feature vectors. Additionally, by sparsifying the model parameters, we further eliminate features weakly related to the prediction results, thereby reducing the model's overconfidence in predictions.

Specifically, we use $P(X; \Omega)$ to denote the probability distribution of GMRF based on the feature data X , and $P(Y|X; \Omega)$ to represent the maximum predict probability based on $P(X; \Omega)$, Ω represents the parameters of the model, Y labels the feature data X . If $P(Y|X; \Omega) \rightarrow 1.0$, X is identified as the label Y ; conversely, if $P(Y|X; \Omega) \rightarrow 0.0$, it is not identified as such. $P(Y|X; \Omega)$ can be considered a general form of a ML model for intrusion detection, which poses little problem when only considering the closed set scenario. However, in practical research, we found that when the model receives unknown feature data X_U , the predicted probability $P(Y|X_U; \Omega)$ often approaches the larger probability, which means that it is identified as a known type. If an unknown attack is identified as a known attack, the harm is relatively minor but still detrimental to our defense analysis work; if it is identified as benign traffic, the damage can be enormous and immeasurable.

We decompose the model overconfidence problem into two stages. In the first stage, we build a multiclass classification model based on the known traffic space (including benign traffic and known attack traffic) and optimize it by minimizing its empirical risk. The specific process can be expressed as follows.

$$\mathcal{R}_1(\Omega) = \frac{1}{N} \left(\sum_{i=1}^N \mathcal{L}(P(Y_i|X_i; \Omega), Y_i) \right), \quad (8)$$

where N denotes the number of sample pairs, $\mathcal{L}(\cdot)$ corresponds to a loss function, e.g., cross-entropy loss for classification.

Clearly, the first stage aims to learn the probability distribution of the GMRF for the characteristic vector X and perform classification tasks within the known space. To further eliminate the impact of weakly correlated features on the prediction results and improve the confidence of the model, we perform parameter sparsification with L_0 regularization in the second stage inspired by [43]. This is achieved by retraining based on the pre-trained model from the first stage. Specifically, we

consider the regularized empirical risk minimization process with regularization L_0 on spurious parameters:

$$\mathcal{R}_2(\Omega) = \frac{1}{N} \left(\sum_{i=1}^N \mathcal{L}(P(Y_i|X_i; \Omega), Y_i) \right) + \lambda \|\Omega\|_0, \quad (9)$$

$$\|\Omega\|_0 = \sum_{j=1}^{|\Omega|} I[\Omega_j \neq 0],$$

where $|\Omega|$ represents the number of parameters, λ is a weighting factor for the regularization of L_0 .

We reduce the prediction probability of the model when encountering unknown attack data in the second stage. Subsequently, by setting a minimum prediction probability threshold, we will distinguish unknown attack types.

B. Practical Solution

To achieve the objectives outlined in Eq. (8) and Eq. (9), we decompose the task into three modules as shown in Fig. 2, specifically: *Pre-Processing*, *GMRF*, and *Sparsification*.

1) *Pre-Processing*: We use CICFlowMeter [44] to parse the raw data, extracting more than 80 features related to network flows. We remove all discrete features (as the experiments showed that they had no impact on model results) and remove a small number of continuous features with many outliers. Ultimately, we applied the Min-Max Scaler [45] to the feature data.

a) *GMRF*: According to Eq. (7), we calculate XX^T , the covariance matrix, which served as input to the model. Then, we model GMRF using a convolutional network [46] and an attention network [47]. The convolutional network is used to implement a *Patch Embedding*, sampling the relational subgraphs (clusters) within the traffic feature space. The attention network is used to learn the edge potential functions. Through these two components, we can realize the GMRF. Additionally, we add a fully connected layer after the GMRF to guide its learning according to Eq. (8) and Eq. (9). Details of *GMRF* are provided in Section IV.

2) *Sparsification*: To better address the issue of model overconfidence, we employ a sparsification method with L_0 regularization [43] for the over-parameterized model to retrain the pre-trained model from the first stage. The calibrated prediction probability distribution of the retrained model can more effectively distinguish unknown attacks. Details of *Sparsification* are provided in Section V.

In general, the workflow in the training phase is as follows. Firstly, the traffic data are processed by the *Pre-Processing* module and normalized into one-dimensional feature vectors. These vectors are then transformed into covariance matrices according to Eq. (7) and fed into the *GMRF* module, where the first training stage is performed using the optimization objective defined in Eq. (8). This stage enables the model to achieve strong detection performance within the known traffic space, although it may also lead to overconfidence when encountering unknown attack traffic. The model then undergoes a second training stage with the *Sparsification* module, whose optimization objective as described in Eq. (9). This step reduces the parameters that contribute to model overconfidence while offering limited practical value, thus improving the reliability of the final model decisions. In

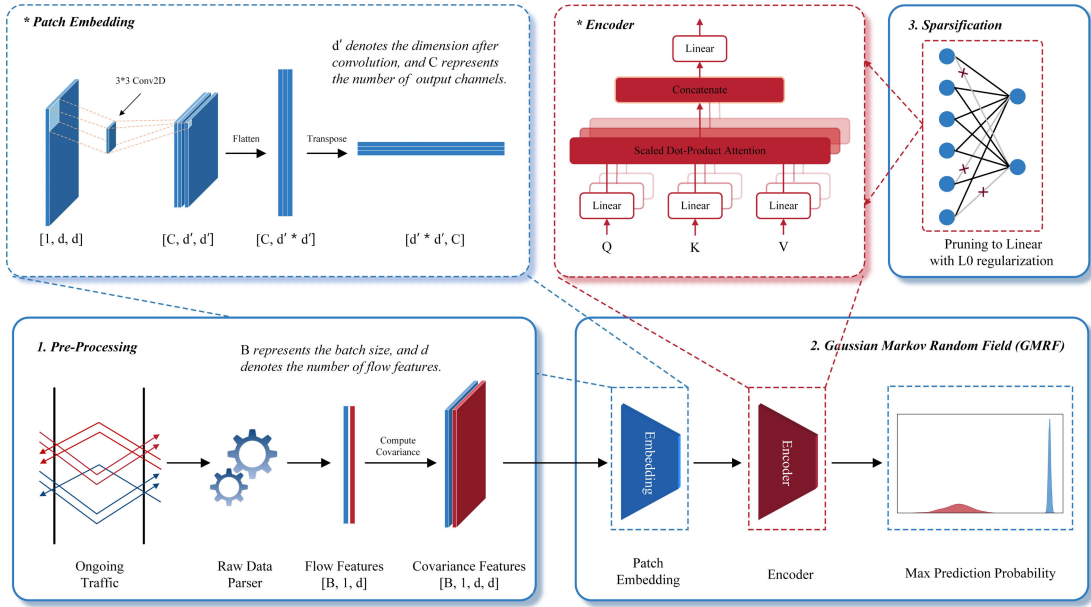


Fig. 2. The overview of our method. It consists of three main modules: **Pre-Processing**, **GMRF**, **Sparsification**. **Pre-Processing** involves extracting feature information from the network flow and computing its covariance matrix for model input. **GMRF** includes a **Patch Embedding** layer and a shallow **Attention Block**. **Patch Embedding** layer flattens the covariance matrix to meet the input requirements of the attention mechanism, while the **Attention Block** performs feature extraction to accomplish the classification task. After **Pre-Processing**, we optimize the model using a **Sparsification** module to remove learning related to weakly relevant features, thereby reducing the model's overconfident predictions.

the inference phase, the workflow only requires the *Pre-Processing* module and the *GMRF* model. Compared with traditional AI-based classification models, ZeroXpert differs in that it determines whether a sample corresponds to an unknown attack based on the output probability distribution of the model.

IV. GMRF

In this section, we provide a detailed explanation of the implementation of the GMRF model. It mainly consists of a *Patch Embedding* layer and an *Encoder* layer, which respectively implement the clusters and edge potential functions in the GMRF.

A. Patch Embedding

Assuming that the input covariance matrix is $X_{cov} \in \mathbb{R}^{1 \times d \times d}$, we perform a 3×3 convolution on it. This operation can extract spatial correlation information within partial feature subgraphs of the matrix, thus capturing the cluster structure of the GMRF model described in Eq. (7). Specifically, it can be expressed as follows.

$$X_{conv} = \text{Conv2D}(X_{cov}, k=3), X_{conv} \in \mathbb{R}^{C \times d' \times d'}, \quad (10)$$

where C denotes the output channels of *Conv2D*.

Then, to meet the input shape requirements of the attention mechanism (*Encoder*), we flatten the last two dimensions of X_{conv} , followed by a matrix transpose. This is expressed as follows:

$$\begin{aligned} X_{fla} &= \text{Flatten}(X_{conv}, -1, -2), X_{fla} \in \mathbb{R}^{C \times (d' \times d')}, \\ X_{trans} &= X_{fla}^T, X_{trans} \in \mathbb{R}^{(d' \times d') \times C}, \end{aligned} \quad (11)$$

where $\text{Flatten}(\cdot)$ refers to *Flatten*.

B. Encoder

The attention mechanism [47] is widely used in tasks such as natural language processing and computer vision, demonstrating strong performance. We utilize an encoder composed of blocks of the attention mechanism to model the edge potential functions in GMRF, aiming to extract spatial relationships between the feature subgraphs. Considering the need for real-time performance in network intrusion detection systems, we employ a shallow neural network to achieve this functionality and remove the residual structure from the attention mechanism blocks. Modified blocks include an attention layer, a feedforward neural network, and two layers of layer normalization [48], as specified below:

$$\begin{aligned} X_{mha} &= \text{MHA}(X_{trans}, X_{trans}, X_{trans}), X_{mha} \in \mathbb{R}^{(d' \times d') \times C}, \\ X_{ln} &= \text{LayerN}(X_{mha}), X_{ln} \in \mathbb{R}^{(d' \times d') \times C}, \\ X_{ffn} &= \text{FFN}(X_{ln}), X_{ffn} \in \mathbb{R}^{(d' \times d') \times C}, \\ X_{out} &= \text{Flatten}(\text{LayerN}(X_{ffn})), X_{out} \in \mathbb{R}^{(d' \times d' \times C)}, \end{aligned} \quad (12)$$

where $\text{MHA}(\cdot)$ denotes *Multi Head Attention* mechanism, $\text{LayerN}(\cdot)$ represents *Layer Normalization*, $\text{FFN}(\cdot)$ stands for *FeedForward Neural Network*.

After the encoding layer, we add a fully connected layer to output the predicted probabilities and perform empirical risk minimization according to Eq. (8) and Eq. (9) during two different training stages.

V. SPARSIFICATION

To further optimize the objective in Eq. (9), due to the non-differentiability and combinatorial nature ($2^{|\theta|}$ possible states) of the parameter vector θ , optimizing under this penalty is computationally infeasible. How can we relax the discreteness

TABLE I
EXPLANATION OF ATTACK TYPES IN THE DATASET

Attack Type	Explanation
FTP-Patator / BruteForce	Dictionary-based brute-force login attempts on FTP services (using Patator tool), avoiding lockout triggers.
SSH-Patator / BruteForce	Dictionary-based brute-force login attempts on SSH services (using Patator tool), avoiding lockout triggers.
DoS-GoldenEye	Opens multiple concurrent connections and sends incomplete HTTP requests to exhaust server resources.
DoS-Slowloris	Keeps many HTTP connections open to exhaust server resources.
DoS-SlowHttptest	Slow-rate HTTP attack using modified Apache benchmarking tool, delaying request headers to exhaust connections.
DoS-Hulk	High-volume HTTP requests to overwhelm web servers.
DDoS	Distributed flood attacks (e.g., SYN/CMP flood) from multiple sources to exhaust network/server resources.
PortScan	Probing a large number of ports to discover open services.
DDoS-HOIC / LOIC-HTTP	Publicly available DDoS tools (LOIC/HOIC) generating high-volume HTTP traffic from multiple sources.
Botnet	Traffic from compromised hosts under centralized control (C&C), often for DDoS, spam, or data exfiltration.
Web-BruteForce	Password guessing attacks targeting web login pages.
Web-XSS	Injection of malicious scripts into web applications.
Infiltration	Unauthorized internal host compromise via malicious payloads.
SQL Injection	Malicious SQL queries to manipulate backend databases.
Heartbleed	Exploits OpenSSL Heartbeat vulnerability to leak sensitive data.

of the L_0 penalty to allow efficient continuous optimization of Eq. (9) while still permitting exact zeros in the parameters?

Following [43], we introduce a binary gate z_j to indicate whether a parameter exists, where the L_0 regularization represents the number of open gates.

$$\Omega_j = \tilde{\Omega}_j z_j, \quad z_j \in \{0, 1\}, \quad \tilde{\Omega}_j \neq 0, \quad \|\Omega\|_0 = \sum_{j=1}^{|\Omega|} z_j. \quad (13)$$

Letting each gate follow a Bernoulli distribution $q(z_j|\pi_j) = \text{Bern}(\pi_j)$, we can reformulate Eq. (9), as follows:

$$\begin{aligned} \mathcal{R}_2(\tilde{\Omega}, \pi) &= \frac{1}{N} \left(\sum_{i=1}^N \mathcal{L} \left(P(Y_i|X_i; \tilde{\Omega} \odot z), Y_i \right) \right) + \lambda \sum_{j=1}^{|\Omega|} \pi_j \\ &= \mathcal{L}_E + \lambda \mathcal{L}_C, \end{aligned} \quad (14)$$

where $\odot$ denotes the product in elemental terms.

Due to the discrete nature of z , it is challenging to optimize Eq. (14) using gradient-based methods. Using a uniform distribution u and a binary concrete random variable s which is distributed in the interval $(0, 1)$, we can reparameterize (“smooth”) z in Eq. (14) as follows:

$$\begin{aligned} u &\sim \mathcal{U}(0, 1), \\ s &= \text{Sigmoid}((\log(u) - \log(1 - u) + \log(\alpha))/\beta), \\ \tilde{s} &= s(\zeta - \gamma) + \gamma, \\ z &= \min(1, \max(0, \tilde{s})), \end{aligned} \quad (15)$$

where α denotes location and β represents temperature, $(\zeta > 1, \gamma < 0)$ are constants to stretch the distribution interval of s .

Consequently, the L_0 complexity loss $\mathcal{L}_C$ of the objective in Eq. (14) can be expressed as:

$$\mathcal{L}_C = \sum_{j=1}^{|\Omega|} \text{Sigmoid} \left(\log(\alpha_j) - \beta \log \left(\frac{-\gamma}{\zeta} \right) \right). \quad (16)$$

VI. EXPERIMENTAL EVALUATION

A. Experiment Setup

1) *Implementation*: We developed the ZeroXpert prototype using Python 3.8.12, and the other modules are built using scikit-learn [49] and PyTorch (2.0.1+cu117) [50].

2) *Implementation*: The system is deployed on a Super-micro server with an 80-core Intel(R) Xeon(R) Gold 6138 CPU, 1 NVIDIA A100-PCIE-40GB, Ubuntu 22.04.4 LTS and 64GB of DRAM (details of our implementation can be found in Appendix B).

3) *Datasets*: We used two publicly available datasets, CICIDS-2017 [40], [41] and CSE-CICIDS-2018 [41], [42], to evaluate the performance of ZeroXpert. Both datasets contain benign traffic and up-to-date samples of common attacks, with data characteristics that closely resemble real-world data (such as in PCAP format). The CICIDS-2017 dataset includes common attack types based on the 2016 McAfee report, such as Web attacks, brute force, DoS, DDoS, Infiltration, Heartbleed vulnerabilities, Botnet and scanning attacks, and the CSE-CICIDS-2018 dataset covers various attack types including brute force, heartbleed, botnet, DoS, DDoS, Web attacks, and internal network penetration. The explanations of the attack abbreviations are provided in Table I.

We used CICFlowMeter [44] to extract network flow features, resulting in more than 80 features. Network flow features are typically indexed using five tuples (source IP, source port, destination IP, destination port, protocol), extracting protocol-related, packet length-related, and time-related features. These flow features can be analyzed without accessing the plaintext content of the packets, making them particularly suitable for encrypted traffic scenarios. Since there is no need to decrypt the traffic, using flow features for detection effectively protects user privacy. Calculating flow features is generally more efficient than content analysis, enabling more real-time intrusion detection. We partition attacks in both datasets into known and unknown categories: benign traffic and known attack data are used for model training, while unknown attack data are excluded from the training process. For known traffic, the data set is randomly divided into the training set and the test set with a 7:3 ratio, and the partitioning and size of the data set are summarized in Table II. Furthermore, considering the size of the data set and the need to reduce computational costs, we performed random sampling on the experimental data set.

4) *Baselines*: We used three state-of-the-art unknown attack detection methods and one traditional machine learning-based method as baselines.

TABLE II
PARTITION OF KNOWN TRAFFIC & UNKNOWN ATTACKS

Datasets	Known	Unknown
CICIDS-2017	Benign (40920)	Botnet (1966)
	FTP-Patator (6705)	Web-BruteForce (1507)
	SSH-Patator (5863)	Web-XSS (652)
	DoS-GoldenEye (7254)	Infiltration (36)
	DoS-Slowloris (2641)	SQL Injection (21)
	DoS-Slowhttptest (4258)	Heartbleed (11)
	DoS-Hulk (17794)	
	DDos (12649)	
	PortScan (14258)	
CICIDS-2018	Benign (19907)	Botnet (14310)
	FTP-BruteForce (19334)	BruteForce-Web (362)
	SSH-BruteForce (18759)	BruteForce-XSS (151)
	DoS-GoldenEye (16603)	SQL Injection (53)
	DoS-Slowloris (10990)	Infiltration (2066)
	DoS-Hulk (13857)	
	DDoS-HOIC (17150)	
	DDoS-LOIC-HTTP (14405)	

- The numbers in parentheses denote the sample sizes used in this experiment.

- *Binary-CIs (machine learning based method)*. We implemented a method based on a combination of random forest [51] and one-class SVM [52], where random forest is used for binary classification between benign and known attacks, while one-class SVM distinguishes between known and unknown attacks.
- *Kitsune@NDSS (reconstruction based method)*. Mirsky et al. [18] employed a stacked auto-encoder neural network to jointly differentiate between benign and malicious traffic patterns.
- *MTH-IDS@IoTJ (signature- or anomaly-based method)*. Yang et al. [26] proposed using a clustering label (CL) k-means model as the initial layer to detect unknown threats. In the second layer of unknown threat detection, they incorporated Bayesian optimization with Gaussian processes and two biased classifiers to optimize the model and reduce the classification errors of the CL-k model.
- *CVAE-EVT@TIFS (reconstruction-based method)*. Yang et al. [15] investigated the reconstruction error between the generated instances, conditioned on labels predicted by a probabilistic discriminative model (known attack detector) and the input instances by training a generative model. They then combined this approach with extreme value theory to detect unknown attacks.
- *CADE@Security (machine learning based method)*. Yang et al. [53] proposed a contrastive learning-based approach that models the distributions of different attack families in the latent space. Using the distance to cluster centroids together with robust statistics, the method can detect distributional drifts and identify zero-day attacks.

5) *Metrics*: We validated the prototype using precision, recall, F1 score, and false positive rate (FPR), as these are widely used in the literature [15], [26].

6) *Hyper-Parameter Selection*: Our PatchEmbedding utilizes a 3×3 convolutional kernel with 1 input channel, 8 output channels, and a padding of 1. In the encoder, we implemented 1 Attention Block configured with 4 heads and a dropout rate of 0.5.

TABLE III
FPR (BENIGN) ON CICIDS-2017 AND CSE-CICIDS-2018 DATASET

Stage	CICIDS-2017	CICIDS-2018
GMRf	1.02%	1.19%
GMRf + Sparsification	1.97%	4.91%

B. Accuracy Evaluation

Fig. 3 summarizes the detection performance of ZeroXpert and presents the results of related ablation experiments. Our ablation study is conducted by sequentially removing the GMRf component and the sparsification process to evaluate their impact. Our evaluation focuses on unknown attacks, and we paid particular attention to the rate of benign traffic misclassified as attack traffic.

1) *Detection Performance of ZeroXpert*: First, we evaluated the recall performance of ZeroXpert on two different datasets. As shown in Fig. 3b, in the CICIDS-2017 data set, ZeroXpert demonstrates excellent recall rates for all known traffic (benign + known attacks), with all attack types exceeding 99% recall, except for the DoS Slowloris attack, which has a recall rate of 82%. Furthermore, the system achieves a recall rate of approximately 57% for unknown attacks. Similarly, as illustrated in Fig. 3e, ZeroXpert also achieves outstanding recall rates on the CSE-CICIDS-2018 dataset, with results ranging from 95% to 99% for known traffic and around 78% for unknown attacks.

Second, we assessed the precision performance of ZeroXpert on two different datasets. As illustrated in Fig. 3a, ZeroXpert achieves a detection precision between 90% and 99% in the CICIDS-2017 dataset, demonstrating excellent performance. Similarly, as shown in Fig. 3d, ZeroXpert performs well in the CSE-CICIDS-2018 data set, with detection precision ranging from 95% to 99% for all types of attacks except benign traffic. Given that attack traffic can be mistaken for benign traffic and that some false positives are inevitable, it is understandable that the precision for detecting benign traffic is slightly lower. The precision in attack traffic detection is a crucial metric in intrusion detection, reflecting the extent to which the model misclassifies attacks. Higher precision indicates fewer benign traffic instances misclassified as attacks. From an impact perspective, a large number of benign traffic misclassified as attacks is particularly problematic, as it could severely disrupt user services.

Third, we evaluated the F1-Score performance of ZeroXpert on two different datasets. As shown in Fig. 3c, ZeroXpert achieves excellent F1 scores in the CICIDS-2017 data set, with all types except for unknown attacks scoring between 90% and 99%. The F1 score for unknown attacks is approximately 70%. Similarly, Fig. 3f illustrates that ZeroXpert performs exceptionally well on the CSE-CICIDS-2018 dataset, with F1 scores exceeding 90% for most types and the F1 score for unknown attacks around 86%. It should be noted that unknown attack data were not included in model training, and achieving such high performance for unknown attack detection without affecting the detection results for known traffic is particularly impressive.

In real-world deployment scenarios, the FPR in benign traffic is a crucial indicator of the practicality of an NIDS.

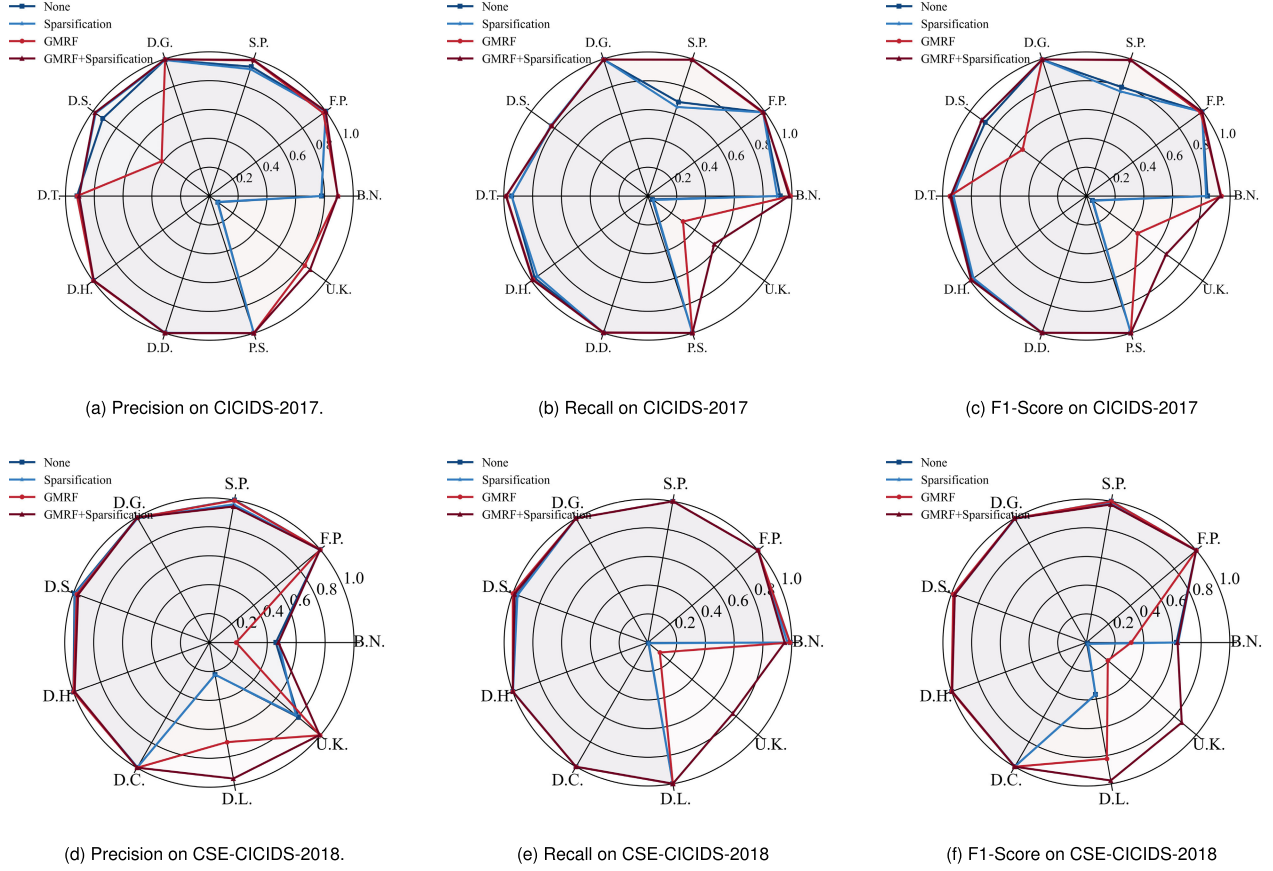


Fig. 3. Performance evaluation of ZeroXpert on different attack types in the CICIDS-2017 and CSE-CICIDS-2018 datasets, along with ablation study results. The abbreviations in the figure are defined as follows: B.N. refers to Benign, F.P. refers to FTP-Patator, S.P. refers to SSH-Patator, D.G. refers to DoS-Goldeneye, D.S. refers to DoS-Slowloris, D.T. refers to DoS-Slowhttptest, D.H. refers to DoS-Hulk, D.D. refers to DDoS, P.S. refers to Portscan, D.C. refers to DDoS-HOIC, D.L. refers to DDoS-LOIC-HTTP, U.K. refers to Unknown attacks.

An excessively high FPR can cause a significant amount of benign traffic to be incorrectly flagged as malicious, potentially disrupting legitimate operations and compromising the usability and stability of the system. As presented in Table III, our method achieved a test FPR of 1.02% in the CICIDS-2017 data set prior to the sparsity pruning process. After pruning and subsequent reconstruction-based retraining, the FPR increased slightly to 1.97%. Similarly, in the CICIDS-2018 dataset, the FPR rose from 1.19% before pruning to 4.91% after pruning. Although the pruning stage led to a moderate increase in FPR, the overall values remained within an acceptable and practical range for deployment.

2) Visual Analysis of Results: We conducted a visual analysis of the detection results to improve the trustworthiness of the predictions made by the proposed ZeroXpert. Specifically, we determined whether a sample is classified as an unknown attack based on whether the model's maximum prediction probability exceeds a predefined trustworthiness threshold. To visually compare the distribution of the maximum prediction probabilities for unknown attacks versus known traffic, we plotted the distribution for all test data. Fig. 4 presents these results: Fig. 4a and Fig. 4b show the maximum prediction probability distributions for ZeroXpert during the pretraining and retraining stages in the CICIDS-2017 dataset, while Fig. 4c and Fig. 4d represent the same for the CSE-CICIDS-2018 dataset.

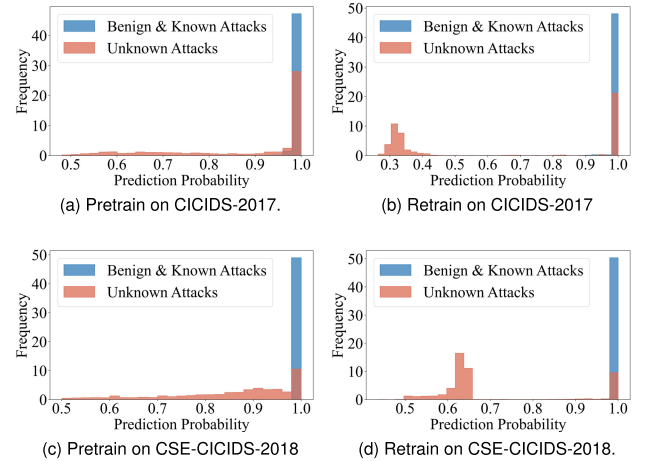


Fig. 4. Distribution of maximum prediction probabilities for ZeroXpert at different training stages.

As shown in Fig. 4a, after pre-training in the CICIDS-2017 dataset, the maximum prediction probabilities for known traffic (including benign traffic and known attacks) in ZeroXpert are mostly concentrated above 0.95. However, a significant portion of the unknown attack traffic also has maximum prediction probabilities exceeding 0.9, indicating that the model is overly confident in detecting this type of traffic. In addition, a small portion of the traffic has maximum prediction

probabilities distributed across various ranges. After applying sparse pruning, as depicted in Fig. 4b, the maximum prediction probabilities for known traffic remain above 0.95, while more than half of the unknown attack traffic has its maximum prediction probabilities concentrated between 0.3 and 0.4. Fig. 4c shows the ZeroXpert pre-training results in the CSE-CICIDS-2018 dataset, where the maximum prediction probabilities for known attack traffic are also higher than 0.95, while those for unknown attack traffic are concentrated between 0.8 and 1.0. After sparse pruning, the maximum prediction probabilities shift to between 0.5 and 0.65. These results demonstrate that by analyzing the differences in probability distributions, it is possible to effectively distinguish the presence of unknown attacks. Sparse pruning helps reduce the influence of features weakly correlated with the model's predictions, thereby lowering the model's overconfidence in unknown attack data and enabling a more effective identification of unknown attack traffic.

3) *Ablation Experiments*: We conducted three ablation experiments on two different datasets, CICIDS-2017 and CSE-CICIDS-2018, to evaluate the impact of GMRF and sparsification on model prediction performance. Initially, regarding recall, when neither GMRF nor Sparsification is applied, the model performs well in recognizing known traffic on both datasets but struggles significantly in detecting unknown attacks (for this experiment, we remove the Patch Embedding, leaving only one Attention Block, as spatial structure extraction is unnecessary). Next, we applied sparsification to the model, which does not lead to an improvement in recall. We then assessed the effect of incorporating GMRF on the prediction results. In the CICIDS-2017 dataset, the model achieves a recall rate of approximately 30% to detect unknown attacks, while in the CSE-CICIDS-2018 dataset, the recall rate for unknown attacks is only about 10%. In comparison, when both GMRF and sparse pruning are applied, the recall rate to detect unknown attacks increases by nearly 30% in the CICIDS-2017 dataset, reaching approximately 57%. The improvement is even more pronounced on the CSE-CICIDS-2018 data set, with an increase of nearly 70%, resulting in a final recall rate of 78%.

Furthermore, we observed a significant improvement in precision when GMRF and sparsification are combined. This suggests that the integration of these two methods effectively removes features weakly correlated with the prediction results, thereby reducing the model's overconfidence and substantially lowering the likelihood of errors. In addition, the F1 score on both data sets also shows superior performance after the combination of GMRF and sparsification.

C. Sensitivity Analysis of Kernel Size

To evaluate the sensitivity of the Patch Embedding module in the ZeroXpert model to convolutional kernel sizes, a series of experiments were conducted on the CICIDS-2017 dataset with varying kernel dimensions. In these experiments, the convolutional kernel size was sequentially varied from 2 to 10, covering a total of nine different sizes with a stride of 1. For each experimental setting, the recall rate and the F1 score were separately assessed for benign traffic and unknown attack traffic. The corresponding experimental results are presented in Fig. 5.

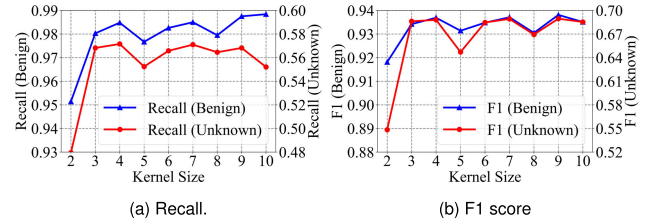


Fig. 5. Sensitivity analysis of Kernel size on CICIDS-2017 dataset.

As shown in Fig. 5a, except for the case where the kernel size is 2, the experimental results for other kernel sizes remain within a stable range. The recall rate for benign traffic is approximately between 0.98 and 0.99, while the recall rate for unknown attack traffic fluctuates slightly within the range of 0.55 to 0.57. Furthermore, as indicated in Fig. 5b, similarly, except when the kernel size is 2, the other experimental results also demonstrate considerable stability. This indicates that the Patch Embedding exhibits favorable sensitivity to kernel size variation, without relying on a specific kernel size to achieve its effect, and highlights the method's desirable stability and robustness. When the kernel size is 2, the performance is relatively poorer, which may be attributed to an insufficient receptive field that fails to capture meaningful patches, thus limiting the formation of discriminative patterns.

D. Comparison With SOTA

To compare detection performance with current research, we selected three existing state-of-the-art (SOTA) methods: Kitsune [18], MTH-IDS [26], CVAE-EVT [15] and CADE [53]. In addition, we deployed a traditional machine learning-based intrusion detection system, Binary-Cls. This system first uses a random forest algorithm to perform binary classification between benign traffic and known attacks, followed by the application of a one-class SVM model to distinguish between known and unknown attacks. Our primary focus is on the detection performance of unknown attacks while ensuring the accuracy of detecting known data (if we solely pursued detection performance for unknown attacks, it can be achieved by adjusting the threshold, but this would significantly decrease the detection accuracy for known data). Therefore, we calculate the precision (Precision-B), recall (Recall-B), and the F1 score (F1 Score-B) for benign traffic, as well as the precision (Precision-U), recall (Recall-U) and the F1 score (F1 Score-U) for unknown attack traffic. The results are presented in Table IV. (Given the instability of the MTH-IDS experimental results, we conducted eight experiments and report the mean values.)

Firstly, we observe the performance of two reconstruction-based methods. CVAE-EVT demonstrates relatively good performance in the CICIDS-2017 data set, achieving a recall rate of 97.66% for benign traffic and 49% for unknown attacks. In addition, it achieves decent levels of precision and an F1 score. In contrast, Kitsune performs moderately, with a recall rate of 50.25% for unknown attacks and only 40.94% for benign traffic, alongside similarly average precision and F1 score results. In comparison, our ZeroXpert shows a recall rate of 98% for benign traffic and 56.82% for unknown attacks on the CICIDS-2017 dataset, surpassing the other two methods.

TABLE IV
COMPARISON OF DIFFERENT METHODS FOR DETECTION OF UNKNOWN ATTACKS

Datasets	Methods	Evaluation Metrics					
		Pre-B (%)	Rec-B (%)	F1-B (%)	Pre-U (%)	Rec-U (%)	F1-U (%)
CICIDS-2017	Binary-Cls	81.65(↓ 7.57)	99.74(↑ 1.71)	89.80(↓ 3.62)	24.40(↓62.21)	18.26(↓38.56)	20.89(↓47.73)
	Kitsune @NDSS	75.40(↓13.82)	40.94(↓57.09)	53.07(↓40.35)	18.71(↓67.90)	50.25(↓ 6.57)	27.26(↓41.36)
	MTH-IDS @IoTJ	79.40(↓ 9.82)	86.27(↓11.76)	82.62(↓10.80)	19.79(↓66.82)	15.94(↓40.88)	17.00(↓51.62)
	CVAE-EVT @TIFS	88.31(↓ 0.91)	97.66(↓ 0.37)	92.75(↓ 0.67)	93.45(↑ 6.84)	49.01(↓ 7.81)	64.30(↓ 4.32)
	CADE @USENIX	79.87(↓ 9.35)	83.42(↓14.61)	81.61(↓11.81)	15.39(↓71.22)	23.15(↓33.67)	18.49(↓50.13)
	ZeroXpert	89.22	98.03	93.42	86.61	56.82	68.62
CICIDS-2018	Binary-Cls	15.46(↓32.29)	99.27(↑ 4.18)	26.75(↓36.83)	1.64(↓97.83)	1.03(↓75.97)	1.26(↓85.54)
	Kitsune @NDSS	16.58(↓31.17)	41.27(↓53.82)	23.66(↓39.92)	85.63(↓13.84)	59.73(↓17.27)	70.37(↓16.43)
	MTH-IDS @IoTJ	31.02(↓16.73)	98.56(↑ 3.47)	44.92(↓18.66)	78.00(↓21.47)	39.93(↓37.07)	44.51(↓42.29)
	CVAE-EVT @TIFS	14.67(↓33.08)	90.11(↓ 4.98)	25.23(↓38.35)	60.00(↓39.47)	2.03(↓74.97)	3.93(↓82.87)
	CADE @Security	50.25(↑ 2.50)	74.32(↓20.77)	59.96(↓ 3.62)	88.77(↓10.70)	86.81(↑ 9.81)	87.78(↑ 0.98)
	ZeroXpert	47.75	95.09	63.58	99.47	77.00	86.80

It also leads in precision and F1 score. In the CSE-CICIDS-2018 dataset, CVAE-EVT performs poorly, with a recall rate of around 90% for benign traffic but only 2% for unknown attacks. Kitsune also shows moderate performance, with a recall rate of 59.73% for unknown attacks and only 41.27% for benign traffic. In contrast, our ZeroXpert maintains an excellent recall rate of 95% for normal traffic in this data set, despite a slight decrease compared to its performance on the CICIDS-2017 data set. However, the recall rate for unknown attacks increases significantly to 77%, with precision and the F1 score far exceeding the other two methods.

Secondly, we focus on methods based on machine learning. In the CICIDS-2017 dataset, MTH-IDS and Binary-Cls achieve recall rates for unknown attacks of approximately 16% and 18%, respectively, with modest recall rates for benign traffic (86% for MTH-IDS). CADE achieves 83.42% recall for benign traffic but only 23.15% for unknown attacks, outperforming MTH-IDS and Binary-Cls in known traffic detection, but still lagging behind ZeroXpert in unknown attack detection. In the CSE-CICIDS-2018 dataset, MTH-IDS and Binary-Cls slightly outperform ZeroXpert in benign traffic detection (98.56% and 99.27%), while CADE achieves a recall of 74.32% for benign traffic and 86.81% for unknown attacks, demonstrating strong performance in unknown attack detection. Despite this, ZeroXpert still maintains superior overall performance, balancing known and unknown traffic detection effectively.

Due to the varying sample sizes of different classes across datasets illustrated in Table II, the precision may differ accordingly. In addition, the complexity of attacks can vary significantly between datasets. For example, some anomalous traffic may be disguised to closely resemble normal traffic. Even for the same attack type, such variations can lead to observable differences in metrics such as the FPR. Nevertheless, we strictly ensure that all comparisons among different baselines are conducted fairly, and each method is evaluated under the same experimental settings and benchmarks.

The effectiveness of ZeroXpert in detecting unknown attacks while maintaining accuracy in detecting known traffic can be attributed to several factors. Firstly, it leverages GMRF to understand the spatial correlations in traffic, thereby preventing the model's predictions from overly depending on local features. This not only enhances the accuracy of known

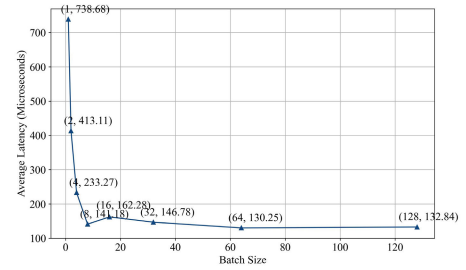


Fig. 6. Detection latency.

traffic detection, but also improves the model's robustness in complex network environments. Secondly, by implementing sparse operations, ZeroXpert effectively removes redundant and unnecessary parameter branches, reducing the impact of noise and anomalous data on predictions. This enables the model to better distinguish between benign and anomalous traffic when facing unknown attacks, thereby reducing misclassifications and lowering the risk of high prediction probabilities. In contrast, several existing SOTA methods do not thoroughly analyze the fundamental reasons for the difficulty in detecting unknown attacks. Although these methods excel in detecting known attacks, they often perform poorly in handling unknown attacks and do not achieve comparable detection effectiveness.

E. Performance Results

1) *Model Size*: The number of model parameters directly determines its memory storage requirements and affects computational complexity and inference time. Given the need for real-time performance in intrusion detection systems, models with a large number of parameters may struggle to meet this requirement. Our proposed ZeroXpert uses a model with a total of 21,936 parameters. This parameter count significantly reduces memory consumption and computational complexity while maintaining model performance, making it effective in resource-constrained environments.

2) *Latency*: We conducted eight independent experiments for each batch size on a low-specification (13th Gen Intel(R) Core(TM) i5-13500H, 32GB DRAM) computer and calculated the average to evaluate the detection latency of ZeroXpert; the

TABLE V
RECOMMENDED HYPER-PARAMETER CONFIGURATION

Group	Hyper-Parameter	Value
Flow Feature	dimension	50
Patch Embedding	in_channels	1
	out_channels	8
	kernel_size	3
	stride	3
	padding	1
Encoder	embedding_size	8
	head_num	4
	ffn_in_features	8
	ffn_hidden_size	16
	ffn_out_features	8
	dropout	0.5
	num_attn_layer	2
Sparsification	α	one-shot random
	β	unstructured pruning
	ζ	0.66
	γ	-0.1

TABLE VI
LATENCY MEASUREMENTS FOR DIFFERENT BATCH SIZE

Batch Size	Experiment							
	# 1	# 2	# 3	# 4	# 5	# 6	# 7	# 8
1	740.82	753.61	755.66	723.33	751.60	717.73	742.45	724.25
2	390.33	432.76	401.43	415.70	421.73	390.41	411.26	441.28
4	240.32	237.26	230.21	237.36	218.17	243.16	214.10	245.58
8	143.60	140.63	148.67	144.61	138.45	135.45	136.71	141.32
16	169.88	174.35	164.47	155.75	160.45	153.03	169.40	150.87
32	170.01	142.12	120.44	153.67	161.85	124.48	165.49	136.21
64	144.07	134.95	140.65	100.35	117.18	147.13	139.76	117.88
128	147.06	125.53	117.57	141.98	144.85	142.40	119.45	123.87

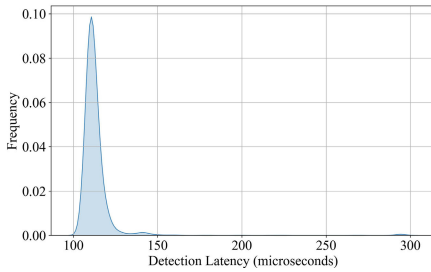


Fig. 7. Distribution of detection latency.

results are shown in Fig. 6 (the raw experimental data can be found in *Appendix B.2*, Table VI)..

Fig. 6 illustrates that as the batch size increases, the detection latency of ZeroXpert decreases from an initial 740 microseconds to approximately 130 microseconds. Furthermore, Fig. 7 presents the density distribution of the latency statistics for all samples (with a batch size of 8). It is evident from this figure that the sample detection latency of ZeroXpert is predominantly within the range of 120 to 130 microseconds. Notably, this evaluation was conducted on a low-spec machine without GPU acceleration. Despite these constraints, ZeroXpert demonstrates a detection latency of approximately 130 microseconds when concurrent processing is considered, meeting the stringent requirements for real-time detection. It is also important to note that our prototype system

is developed in Python. Although Python offers ease of use and flexibility, it may not be the most optimal for performance in computation-intensive tasks. Therefore, implementing the system in a higher-level language like C++ could potentially further reduce the detection latency, thereby enhancing the overall performance and feasibility of the system in practical applications.

VII. RELATED WORK

In this section, we summarize previous related research efforts, with detailed information provided as follows.

A. Machine Learning Based NIDS

Machine learning-based network intrusion detection methods have shown superior performance compared to traditional signature-based methods, particularly in the identification of unknown attacks [3], [54]. Ahmad et al. [30] compared the research methods and performance of traditional machine learning algorithms (such as the support vector machine (SVM) and decision trees) in the detection of zero-day attacks. Fu et al. [25] proposed a real-time malicious traffic detection system based on frequency domain analysis, which achieved accurate and robust network attack detection through feature extraction and automatic parameter selection, combined with statistical clustering algorithms. Yang et al. [53] introduced a contrastive learning-based method that models the distributions of different attack families in latent space, using distances to cluster centroids and robust statistics (e.g., MAD) to detect distributional drift and identify zero-day attacks. Yang et al. [26] introduced a multilayer hybrid intrusion detection system named MTH-IDS to detect known and unknown network attacks in vehicular networks. This system combined signature-based and anomaly-based detection methods and optimized model performance through machine learning algorithms. However, traditional machine learning methods exhibited limited generalization capability and failed to clearly distinguish between unknown and known attacks. Although Yang et al. [26] attempted to separate these categories, the false positive rate (normal traffic being misclassified as attack traffic) became uncontrollable.

Transfer learning has also shown promising applications in intrusion detection. For example, Sameera and Shashi [24] proposed a deep transductive transfer learning framework for zero-day attack detection. This framework aligns the source and target domains into a common latent space through manifold alignment, thereby avoiding discrepancies in feature spaces and probability distributions. It generates soft labels using cluster correspondence programs to address the scarcity of labeled instances in the target domain and employs deep neural networks (DNNs) to build a binary classifier for zero-day attack detection. Taghiyarrenani et al. [21] introduced a knowledge transfer method based on transfer learning to better handle unknown attacks. Furthermore, Kim et al. [22] suggested using transfer deep convolutional generative adversarial networks (tDCGANs) to generate pseudomalevolent traffic based on known attack traffic, thus enhancing zero-day attack detection performance through data set enhancement. Few-shot learning [55] and zero-shot learning [56] are becoming increasingly common in intrusion detection. For example, Zhang et al. [57] introduced a zero-shot learning method based

on sparse autoencoders to detect unknown attacks. Yang et al. [58] proposed a mechanism for training models using a few-shot learning to acquire general knowledge about attacks rather than specific knowledge about individual categories, thereby enhancing the detection of unknown attacks. However, while these methods demonstrated good generalization capabilities, they still failed to explicitly distinguish between known and unknown attacks, which is crucial for defensive purposes.

B. Reconstruction Based Zero-Day Detection

In recent years, reconstruction-based methods to address the challenge of unknown attack detection have gained increasing attention. The primary idea involves first auto-encoding the known traffic and then reconstructing the encoded vectors. Since unknown attack traffic is not included in the training process, it tends to exhibit higher reconstruction errors, making it easier to identify as anomalous [59]. For example, Hindy et al. [16] proposed using autoencoders with a reconstruction error threshold to detect zero-day attacks. Zha et al. [10] suggested applying a sigmoid kernel transformation to map traffic from a low-dimensional space to a high-dimensional space, thus amplifying the differences between unknown attacks and known traffic, and then using an encoder-decoder framework for detection. Yang et al. [15] explored the training of a generative model to examine the deviation between the generated instances conditioned on the labels predicted by a probabilistic discriminative model (a known attack detector) and the input instances. Additionally, they combined extreme value theory to detect unknown attacks. Tang et al. [17] recommended employing an encoder-decoder recurrent neural network to train a self-translation machine, which captures the syntactic and semantic patterns of benign requests while marking incomprehensible requests as attacks. Furthermore, Mirsky et al. [18] introduced a stacked autoencoder neural network designed to jointly distinguish between normal and anomalous traffic patterns.

Reconstruction-based methods have shown promise in the field of unknown attack detection. These approaches not only maintain effective detection performance, but also allow for the adjustment of reconstruction error thresholds to control the false positive rate (i.e., misclassifying normal traffic as attack traffic), which is crucial. However, most of these methods rely on cosine similarity or Euclidean distance to measure the reconstruction error. Given the characteristics of network traffic data (where, after normalization, most feature values are close to 0 and a few are close to 1, as shown in Fig. 1), these metrics tend to focus more on changes in larger values while neglecting smaller ones, making the reconstruction process relatively straightforward. Consequently, these methods are subject to certain limitations. We provide a theoretical analysis to support this point (see Appendix A).

VIII. DISCUSSION

In this section, we discuss our proposed method further and explore potential directions for future work.

A. Adaptation to Dynamic Network Environments

In real-world deployments, network environments are inherently dynamic, where attack patterns continuously evolve and may deviate from previously observed distributions. Rather

than relying on frequent model updates or explicit online learning, ZeroXpert is designed to adapt to such dynamics through robust uncertainty-aware detection. By modeling the spatial relationships among the traffic features using GMRF, ZeroXpert captures stable structural dependencies within the known traffic space. The subsequent sparsification stage further suppresses weakly relevant or spurious features, reducing the model's tendency to produce overconfident predictions. As a result, when encountering traffic generated by evolving or previously unseen attacks, the model typically yields lower prediction confidence instead of forcibly assigning them to known classes. This confidence-aware decision mechanism allows ZeroXpert to effectively flag evolving attack patterns as unknown without immediate retraining, making it well suited for deployment in dynamic network environments where traffic characteristics and attack behaviors change over time.

B. Potential Directions for Future Work

ZeroXpert is designed to robustly identify evolving and previously unseen attacks without frequent retraining. Meanwhile, adapting to long-term distributional shifts and continuously emerging attack patterns is also an important direction for future research. One promising extension is to incorporate lightweight drift detection mechanisms that monitor changes in the statistical properties of incoming traffic or the confidence distribution of the model, thus triggering adaptive responses only when significant data drift is detected. In addition, future work may explore selective or incremental model updates, where newly observed unknown samples with high confidence can be gradually incorporated into the known traffic space without retraining the entire model from scratch. Another potential direction lies in combining ZeroXpert with generative or replay-based techniques to mitigate catastrophic forgetting when adapting to new attack types while maintaining stable performance on previously learned patterns. Finally, investigating theoretically grounded calibration strategies under data drift could further enhance the reliability of confidence-based unknown attack detection in highly dynamic network environments.

Meanwhile, data imbalance is an important issue that deserves careful consideration in network traffic detection, as the number of benign traffic samples typically far exceeds that of attack traffic. Existing studies have explored mitigation strategies such as employing generative models [60] to synthesize minority-class samples to improve class balance. However, these approaches also suffer from inherent limitations, including the potential risk of data poisoning. Developing a novel sampling strategy therefore represents a promising research direction. Unlike random downsampling methods, which reduce the majority class at the cost of losing informative samples, an effective solution to traffic imbalance should aim to reduce sample redundancy while preserving critical information. Such a strategy would have the additional benefit of significantly reducing model training time. These considerations constitute an important part of our future research agenda.

IX. CONCLUSION

In this study, we propose a method based on a Gaussian Markov Random Field to capture the spatial feature

distribution of network traffic, effectively modeling the relationships between local features. GMRF also better satisfies the conditional independence assumption. Building on this, we apply sparsification to the model by pruning weakly correlated parameters, reducing the prediction probability of unknown attack traffic. This creates a significant gap between the prediction probabilities of known and unknown traffic, enabling the effective detection of unknown attacks. As a result, ZeroXpert not only addresses the issue of model overfitting, but also mitigates the problem of overconfident predictions when faced with unknown attack traffic. We evaluated the performance of our approach on two public datasets, CICIDS-2017 and CSE-CICIDS-2018, through both detection performance evaluation and ablation experiments. The results show that our method achieves recall rates of 57% and 77% for unknown attacks while maintaining a recall rate above 95% for known traffic, significantly outperforming several existing SOTA methods. Additionally, performance tests on a low-spec machine demonstrate that the system maintains low detection latency, making it well-suited for real-time requirement.

APPENDIX

A. Limitations of Reconstruction-Based Methods

In this section, we will theoretically elaborate on the potential shortcomings and limitations of unknown attack detection methods based on encoder-decoder architecture [15], [16], [17], [18].

The main idea of the unknown attack detection method based on the encoder-decoder architecture is to first encode the known feature vectors, then decode the encoded semantic vectors, and finally determine the presence of unknown attacks based on the reconstruction error. The calculation of the reconstruction error mainly uses *Cosine Similarity* or *Euclidean Distance*. However, we believe that methods such as cosine similarity and Euclidean distance have theoretical flaws in this direction: *they tend to ignore features with relatively small values, while features with larger values have a more significant impact on the final result (e.g. cosine similarity).*

Assume there are two feature vectors X_1, X_2 , each with m larger features and n smaller features, denoted as $X_1 = (x_{1,1}, x_{1,2}, \dots, x_{1,m}, x_{1,m+1}, x_{1,m+2}, \dots, x_{1,m+n})$ and $X_2 = (x_{2,1}, x_{2,2}, \dots, x_{2,m}, x_{2,m+1}, x_{2,m+2}, \dots, x_{2,m+n})$. The cosine similarity $\cos(X_1, X_2)$ is expressed as:

$$\frac{\sum_{i=1}^m x_{1,i}x_{2,i} + \sum_{j=m+1}^{m+n} x_{1,j}x_{2,j}}{\sqrt{\sum_{i=1}^m x_{1,i}^2 + \sum_{j=m+1}^{m+n} x_{1,j}^2} \sqrt{\sum_{i=1}^m x_{2,i}^2 + \sum_{j=m+1}^{m+n} x_{2,j}^2}} \quad (17)$$

After normalization, all feature values are within the range of $[0, 1]$. For larger value features $x_{1,i}$ and $x_{2,i}$ ($i \in \{1, 2, \dots, m\}$), we assume $x_{1,i} \rightarrow 1, x_{2,i} \rightarrow 1$, and for smaller value features $x_{1,m+j}$ and $x_{2,m+j}$ ($j \in \{1, 2, \dots, n\}$), we assume $x_{1,m+j} \rightarrow 0, x_{2,m+j} \rightarrow 0$. Therefore,

$$\sum_{i=1}^m x_{1,i}x_{2,i} \rightarrow 1, \sum_{j=m+1}^{m+n} x_{1,j}x_{2,j} \rightarrow 0$$

$$\begin{aligned} \sum_{i=1}^m x_{1,i}^2 &\rightarrow 1, \sum_{j=m+1}^{m+n} x_{1,j}^2 \rightarrow 0 \\ \sum_{i=1}^m x_{2,i}^2 &\rightarrow 1, \sum_{j=m+1}^{m+n} x_{2,j}^2 \rightarrow 0 \end{aligned} \quad (18)$$

By Eq. (17) and Eq. (18), we obtain the following.

$$\cos(X_1, X_2) = \frac{\sum_{i=1}^m x_{1,i}x_{2,i}}{\sqrt{\sum_{i=1}^m x_{1,i}^2} \sqrt{\sum_{i=1}^m x_{2,i}^2}} \quad (19)$$

Combining the characteristics of the network flow feature values, most of the feature values do not follow a normal distribution. Typically, a feature value's distribution contains a small number of extremely large values, while the majority are distributed as smaller values (for example, flow duration, where very few flows last for an extended period, while most end within a relatively short time). We have confirmed this phenomenon in several existing datasets [40], [41], [42], [61], which fully supports our assumption. According to Eq. (19), we can see that the reconstruction error depends on the few features with larger values, making the reconstruction less challenging and presenting the "illusion" that unknown attack traffic is highly similar to known traffic.

Similarly, the Euclidean distance is also more sensitive to large feature values and less sensitive to small feature values. This indicates that the Euclidean distance calculation, in cases where feature value differences are substantial, tends to reflect the differences in large feature values more prominently, thereby neglecting the impact of small feature values.

B. Details of Implementations

1) *Detail of Recommended Hyper-Parameter*: Table V shows the hyperparameters used in ZeroXpert and the recommended values.

In the sparsification process, two important hyperparameters ζ and γ are required to satisfy $\zeta < 0.0$ and $\gamma > 1.0$. This design ensures that, after the hard-thresholding operation $z = \min(1, \max(0, \tilde{s}))$, the model can still produce exact 0 and 1 values rather than only approximate ones. We recommend setting $\zeta = -0.1$ and $\gamma = 1.1$, which are empirically chosen [43]. If ζ becomes too negative (e.g., -10), the sigmoid/logistic transformation will push the gating probability excessively toward 0 or 1, damaging the smoothness of the gradient; if ζ is too close to 0, the model will be unable to produce exact zeros. The parameter γ plays a symmetric role in controlling the upper bound. In practice, a small outward stretch (around ± 0.1) is sufficient to allow some samples to fall outside the interval $[0, 1]$, allowing exact 0/1 values after hard thresholding.

Furthermore, in relation to the hyperparameter β , it controls the smoothness of the Concrete distribution. A larger β makes the distribution closer to Bernoulli, whereas a smaller β yields smoother relaxations with a lower gradient variance but a higher bias. We set $\beta = 0.66$, which has been shown in previous work [62] to provide a good balance between training stability and the desired level of sparsity.

2) *Raw Data of Latency Experiments*: Table VI illustrates the raw data from eight independent experiments for the performance test (latency) described in Section VI-E.

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